

LAPORAN TUGAS ALGORITMA PEMROGRAMAN

PEKAN 8



Oleh :

MUHAMMAD ARIF

NIM 2511532017

MATA KULIAH ALGORITMA PEMROGRAMAN

DOSEN PENGAMPU : DR. WAHYUDI, S.T, M.T

FAKULTAS TEKNOLOGI INFORMASI

DEPARTEMEN INFORMATIKA

UNIVERSITAS ANDALAS

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KODE PROGRAM

```
package pekan8_2511532017;
```

```
import java.awt.BorderLayout;
```

```
import java.awt.EventQueue;
```

```
import javax.swing.JOptionPane;
```

```
import javax.swing.JFrame;
```

```
import javax.swing.JPanel;
```

```
import javax.swing.border.EmptyBorder;
```

```
import javax.swing.JLabel;
```

```
import java.awt.Font;
```

```
import javax.swing.JTextField;
```

```
import javax.swing.JComboBox;
```

```
import javax.swing.JButton;
```

```
import javax.swing.DefaultComboBoxModel;
```

```
import javax.swing.SwingConstants;
```

```
import java.awt.event.ActionListener;
```

```
import java.awt.event.ActionEvent;
```

```
public class OperatorAssignmentGUI_2511532017 extends JFrame {
```

```
    private int nilaiBill; //menyimpan nilai Bill yang terus berubah
```

```
    private boolean sudahDiset = false; //kondisi bill masih bisa
    diinputkan
```

```
    private static final long serialVersionUID = 1L;
```

```
    private JPanel contentPane;
```

```
    private JTextField Billtxt;
```

```
    private JTextField Bil2txt;
```

```
    private JTextField OutputHasil;
```

```
    private void pesanPeringatan (String pesan) {
```

```
        JOptionPane.showMessageDialog(this, pesan, "Peringatan",
        JOptionPane.WARNING_MESSAGE);
```

```
    }
```

```
    private void pesanError (String pesan) {
```

```
        JOptionPane.showMessageDialog(this, pesan, "error",
        JOptionPane.ERROR_MESSAGE);
```

```
    }
```

```
    /**
```

```
     * Launch the application.
```

```
     */
```

```
    public static void main(String[] args) {
```

```
        EventQueue.invokeLater(new Runnable() {
```

```
            public void run() {
```

```

        try {
            OperatorAssignmentGUI_2511532017
frame = new OperatorAssignmentGUI_2511532017();
            frame.setVisible(true);
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
});
}

/**
 * Create the frame.
 */
public OperatorAssignmentGUI_2511532017() {
    setResizable(false);
    setTitle("OPERATOR ASSIGNMENT ");
    setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    setBounds(100, 100, 450, 300);
    contentPane = new JPanel();
    contentPane.setBorder(new EmptyBorder(5, 5, 5, 5));
    setContentPane(contentPane);

```

```
contentPane.setLayout(null);
```

```
JLabel lblNewLabel = new JLabel("OPERATOR  
ARITMATIKA");
```

```
lblNewLabel.setFont(new Font("Times New Roman",  
Font.BOLD, 12));
```

```
lblNewLabel.setBounds(142, 10, 194, 29);
```

```
contentPane.add(lblNewLabel);
```

```
JLabel Bil1 = new JLabel("Bilangan 1");
```

```
Bil1.setFont(new Font("Times New Roman", Font.BOLD,  
12));
```

```
Bil1.setBounds(21, 68, 72, 29);
```

```
contentPane.add(Bil1);
```

```
JLabel Bil2 = new JLabel("Bilangan 2");
```

```
Bil2.setFont(new Font("Times New Roman", Font.BOLD,  
12));
```

```
Bil2.setBounds(21, 107, 72, 29);
```

```
contentPane.add(Bil2);
```

```
JLabel Operator = new JLabel("Operator");
```

```
Operator.setFont(new Font("Times New Roman",  
Font.BOLD, 12));
```

```
Operator.setBounds(21, 160, 75, 26);
```

```
contentPane.add(Operator);
```

```
JLabel Hasil = new JLabel("Hasil");
```

```
Hasil.setFont(new Font("Times New Roman", Font.BOLD,  
12));
```

```
Hasil.setBounds(21, 211, 75, 24);
```

```
contentPane.add(Hasil);
```

```
Billtxt = new JTextField();
```

```
Billtxt.setBounds(93, 72, 96, 18);
```

```
contentPane.add(Billtxt);
```

```
Billtxt.setColumns(10);
```

```
Bil2txt = new JTextField();
```

```
Bil2txt.setColumns(10);
```

```
Bil2txt.setBounds(93, 111, 96, 18);
```

```
contentPane.add(Bil2txt);
```

```
JComboBox assignmentCMB = new JComboBox();
```

```
assignmentCMB.setFont(new Font("Times New Roman",  
Font.BOLD, 11));
```

```
assignmentCMB.setModel(new  
DefaultComboBoxModel(new String[] { "+=", "-=", "*=", "/=", "%=" }));  
assignmentCMB.setBounds(93, 162, 57, 20);  
contentPane.add(assignmentCMB);
```

```
JButton btnProses = new JButton("proses");  
btnProses.addActionListener(new ActionListener() {  
    public void actionPerformed(ActionEvent e) {  
  
        if (!sudahDiset && Bil1txt.getText().trim().isEmpty())  
        {  
            pesanPeringatan("Silahkan Masukkan Bilangan 1");  
            return;  
        }  
        if (Bil2txt.getText().trim().isEmpty()) {  
            pesanPeringatan("Silahkan Masukkan Bilangan 2");  
            return;  
        }  
  
        try {  
            int b = Integer.parseInt(Bil2txt.getText());  
  
            // Ambil nilai awal Bil1 hanya sekali
```

```

        if (!sudahDiset) {
            nilaiBill = Integer.parseInt(Billtxt.getText());
            Billtxt.setEnabled(false);
            sudahDiset = true;      // bill sudah tidak dapat
diubah lagi
        }

// Ambil pilihan operator
int op = assignmentCMB.getSelectedIndex();

if (op == 0) {
    nilaiBill += b;
} else if (op == 1) {
    nilaiBill -= b;
} else if (op == 2) {
    nilaiBill *= b;
} else if (op == 3) {
    if (b == 0) {
        pesanPeringatan("Bilangan 2 tidak boleh nol
untuk pembagian!");
        return;
    }
    nilaiBill /= b;
}

```

```

        } else if (op == 4) {
            if (b == 0) {
                pesanPeringatan("Bilangan 2 tidak boleh nol
untuk modulus!");

                return;
            }
            nilaiBill %= b;

        }
        Billtxt.setText(String.valueOf(nilaiBill));

        OutputHasil.setText(String.valueOf(nilaiBill));

    } catch (NumberFormatException ex) {
        pesanError("Input harus berupa angka bulat!");
    }
}

});

btnProses.setFont(new Font("Times New Roman",
Font.BOLD, 12));

btnProses.setBounds(180, 162, 84, 20);

contentPane.add(btnProses);

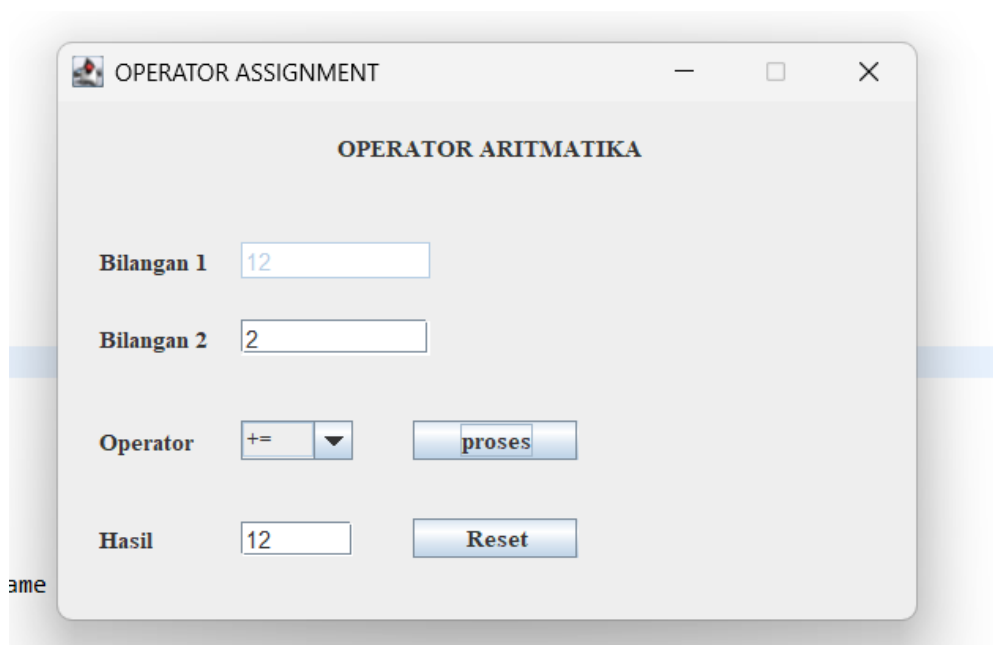
```

```
OutputHasil = new JTextField();  
OutputHasil.setColumns(10);  
OutputHasil.setBounds(93, 213, 57, 18);  
contentPane.add(OutputHasil);
```

```
JBUTTON btnReset = new JBUTTON("Reset");  
btnReset.setFont(new Font("Times New Roman",  
Font.BOLD, 12));  
btnReset.setBounds(180, 212, 84, 20);  
btnReset.addActionListener(new ActionListener() {  
    public void actionPerformed(ActionEvent e) {  
        // Reset semua  
        Bill1txt.setText("");  
        Bill2txt.setText("");  
        Bill1txt.setEnabled(true);  
        OutputHasil.setText("");  
        sudahDiset = false;  
        nilaiBill = 0;  
        // kursor Fokus ke Bill lagi  
        Bill1txt.requestFocus();  
    }  
});
```

```
        contentPane.add(btnReset);  
  
    }  
}
```

OUTPUT



ne

OPERATOR ASSIGNMENT

OPERATOR ARITMATIKA

Bilangan 1

14

Bilangan 2

2

Operator

+=

▼

proses

Hasil

14

Reset

me

OPERATOR ASSIGNMENT

OPERATOR ARITMATIKA

Bilangan 1

12

Bilangan 2

2

Operator

-=

▼

proses

Hasil

12

Reset

me

 OPERATOR ASSIGNMENT

OPERATOR ARITMATIKA

Bilangan 1

24

Bilangan 2

2

Operator

*/=

▼

proses

Hasil

24

Reset

me

 OPERATOR ASSIGNMENT

OPERATOR ARITMATIKA

Bilangan 1

12

Bilangan 2

2

Operator

/=

▼

proses

Hasil

12

Reset

ame

OPERATOR ASSIGNMENT

OPERATOR ARITMATIKA

Bilangan 1

0

Bilangan 2

2

Operator

%=

▼

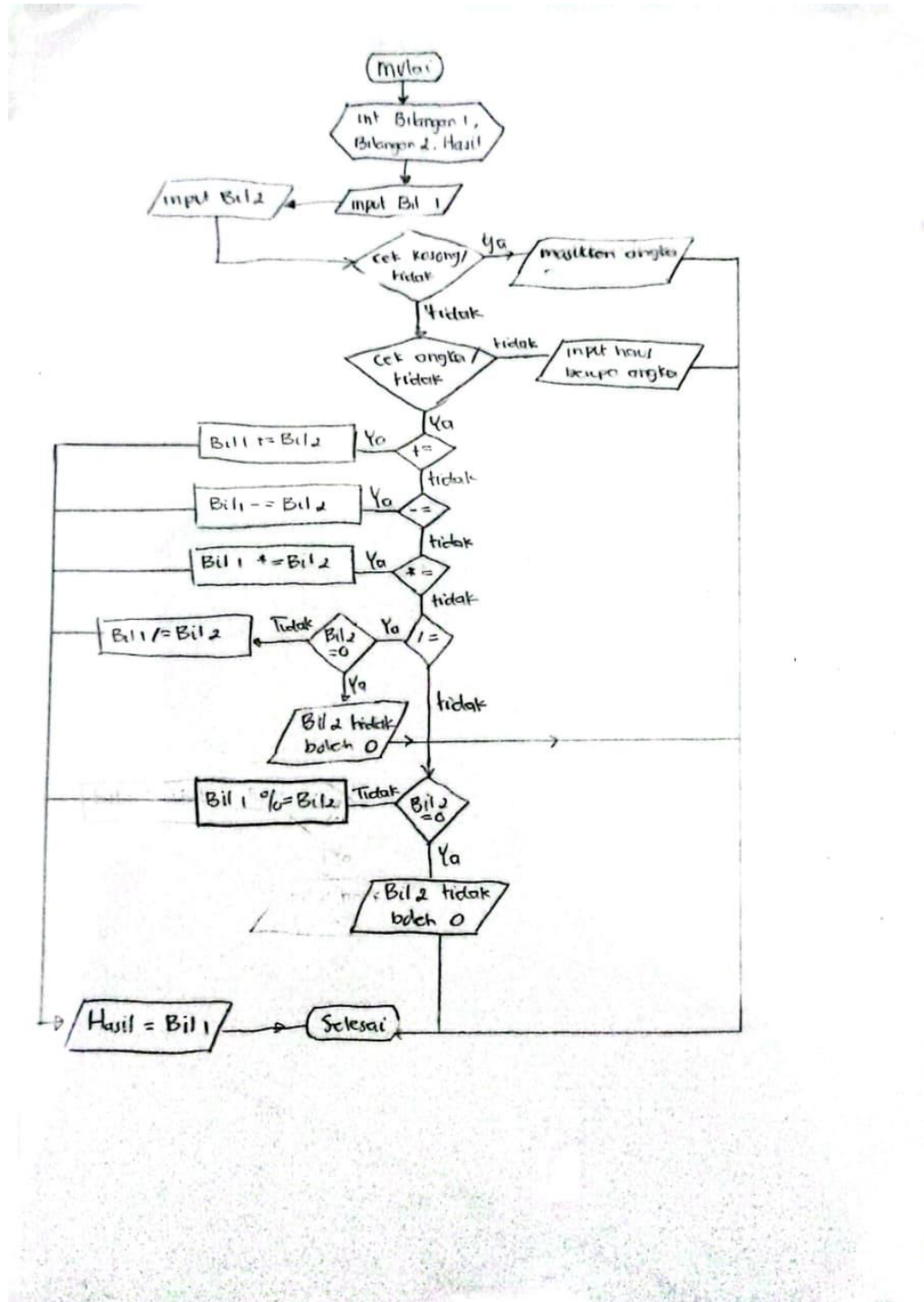
proses

Hasil

0

Reset

FLOWCHART



PSEUDOCODE

Judul Operasi Assignment GUI
Deklarasi Bil1,bil2 ← integer Hasil ←string
Pseudocode Mulai Input bil1 Input bil2 If (bil1 = kosong) Ouput (“silahkan masukkan bilangan 1”) Endif If (bil2= kosong) Output (“silahkan masukkan bilangan 2”) Endif If (bil1 & bil2 bukan angka) Output (“input harus berupa angka “) Endif If (+=) then Bill=bil1+bil2 Else if (-=) then Bill = bil1 – bil2 Else if (*=) then Bill = bil1 * bil2 Else if (/=) then If (bil2=0) Output (“bilangan 2 tidak boleh 0”) else Bill + bil1 / bil2 endif Else (%=) then If (bil2=0) Output (“bilangan 2 tidak boleh 0”) Bill = bil1 % bil2 Endif Endif

Hasil ← bil1
Selesai